Discussion

From Labor to Intermediates: Firm Growth, Input Substitution, and Monopsony

(Mertens-Schoefer)

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Background and Motivation

- Labor shares (LS) vary over time, across+within industry, and across firms

  (e.g. Elsby-Hobijn-Sahin 2013, Karabarbounis-Neiman 2014, Hubmer 2023, Autor-Dorn-Katz-Patterson-Van Reenen 2020, Kehrig-Vincent 2021, Hubmer-Restrepo 2024)

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- Key difficulty: data and methods usually does not allow for arbitrary heterogeneity both in market power and technology

Summary of Approach

- Data: Firm-year panel data, German manuf. 1995–2017
- Estimates translog prod function $\Rightarrow \theta^X = \frac{\partial \ln Q}{\partial \ln X}$ for $X = \{L, K, M\}$ allowed to vary at firm-year level as function of input levels
- Given recovered elasticities $\theta^X_{it} = \theta^X(K_{it}, L_{it}, M_{it})$, correlate with firm growth.
- With additional structure, recover relative shadow price of labor, and labor supply elasticity, from estimated production function and observed factor shares

Finding 1: Growing Firms Substitute Intermediates for Labor

- Growing firms exhibit
  - declining θ^L and θ^K, more so at shorter horizon
  - increasing θ^M, more so at shorter horizon
- Firms expand by relying on intermediates
  - Rejects benchmark of Cobb-Douglas production
  - Labor-intermediate elasticity $\sigma > 1$: L and M are gross substitutes
- Show transparently using descriptive evidence (quantities, factor shares) and (careful!) production function estimation
  - Great data (p and q!) and execution (c.f. Bond-Hashemi-Kaplan-Zoch 2021)
  - Results similar for any firm growth (OLS) and firm growth induced by demand shocks (IV) $\rightarrow$ growing firms move in similar direction in input space

$\Rightarrow$ very well done and convinced me!

Finding 2: Growing Firms' Shadow Price of Labor Increases

- Given homothetic prod fun, $\sigma > 0$ and growing firms' $\frac{L}{M} \downarrow$
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- Interpret as upward-sloping firm-specific labor supply curve
  - Could also be (dynamic) adjustment costs (e.g. Chan-Mattana-Salgado-Xu 2023)
- Assuming flexible M, back out labor supply elasticity
  - $\frac{d \ln L}{d \ln w}$ estimated 2.0 (IV, 1y) to 5.2 (IV, 4y)

$\Rightarrow$ plausible explanation but not fully convinced yet

Finding 3: Aggregate Implications for Declining Labor Share

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- Macro: Would need to close model to think about structural interpretation
  - in terms of national income, is declining labor share going to machines or profits?
  - input-output network (US: Hubmer 2023), international trade...

$\Rightarrow$ important finding, but interpretation outside of scope of paper

Suggestions

1. Stick to same class of prod fun throughout
   - Right now estimate translog prod fun (non-homothetic, $RTS \uparrow$)
   - Interpret results (back out σ, $\frac{d \ln L}{d \ln W}$) via nested CES (homothetic, $RTS = 1$)
   - Theory as if NO restriction on prod fun required (Hicks vs Morishima elast.)

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2. Rule out non-homothetic prod fun driving subs from L to M
   - Robustness using non-homothetic CES prod fun: reg

$$\Delta \ln \frac{\theta_{it}^{L}}{\theta_{it}^{M}} = \frac{\sigma - 1}{\sigma} \Delta \ln \frac{L_{it}}{M_{it}} + \eta \frac{\sigma - 1}{\sigma} Q_{it}$$

   - Argue that since $\hat{\eta} \approx 0$, non-homotheticity not present
   - But θ_{it}^{X} is a function of inputs (L_{it}, M_{it}, K_{it}), controlling for inputs no variation left?

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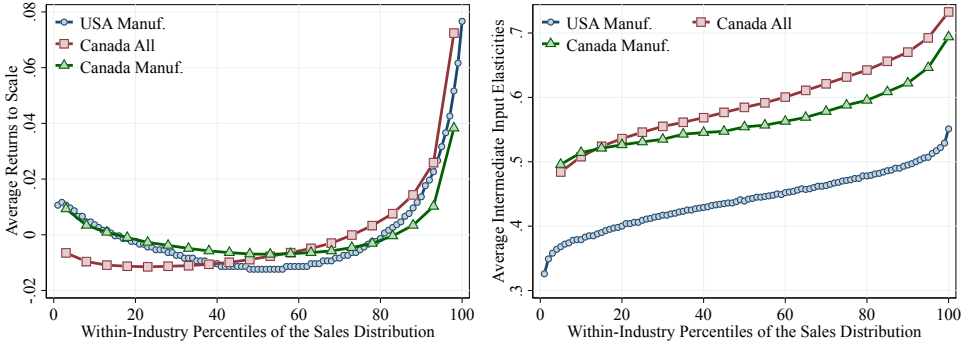
3. Dig into composition of M to understand mechanism
   - coming from materials and energy; merchandise; subcontracted work (Table A.9)
   - quite a different story whether task ultimately performed by labor or machine

Related Findings in Canada and US-Manufacturing



(a) Returns To Scale $= \theta^L + \theta^K + \theta^M$ by Size **(b)** θ^M by Size

- Chan-Hong-Hubmer-Ozkan-Salgado (2024): larger firms have higher RTS, θ^M, lower θ^L, θ^K
- Differences: Focus on cross-section, using non-parametric prod fun estimation

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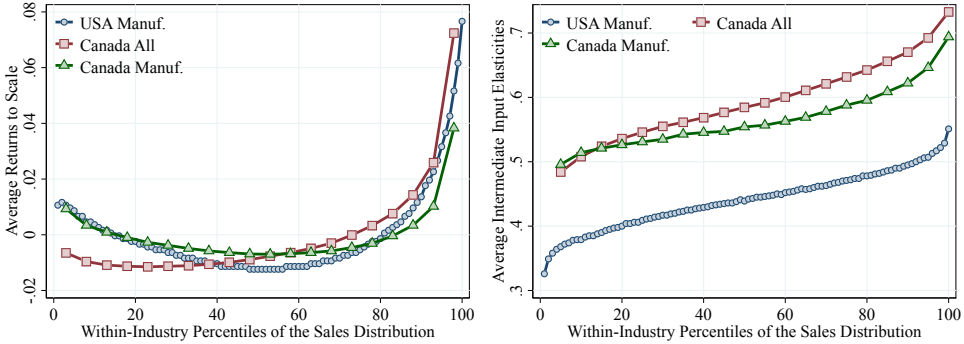

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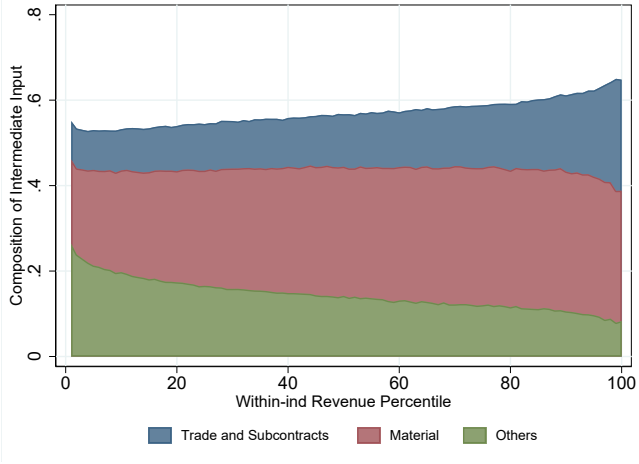
- Differences: Focus on cross-section, using non-parametric prod fun estimation
- Now sketch possible micro-foundation: Outsourcing, no market power, non-homothetic prod fun
  - Suggestion: can reject this and other purely technological explanations?

Motivating Fact: Composition of Intermediate Input Spending



Canadian cross-section: higher intermediate input share driven entirely driven by trade and subcontract labor (not: materials, electricity...)

Propose task-based prod fun with endogenous outsourcing

Possible Micro Foundation: Outsourcing

- Key assumption: each task $j \in [0,1]$ can be produced
  - either in-house with $k_i(j)$ and $\ell_i(j)$, span of control $\eta \in (0,1)$
  - or outsourced purchasing intermediate inputs $m_i(j)$, at CRTS
- firm output y_i:

$$\ln y_i = \ln z_i^R + \int_0^1 \ln x_i(j)dj$$

$$\text{where} \quad x_i(j) = \left(\underbrace{k_i(j)^\alpha \ell_i(j)^{1-\alpha}}_{\text{value added bundle, cost } c_v} \right)^\eta + z^m(j)m_i(j)$$

Possible Micro Foundation: Outsourcing

- task allocation characterized by outsourcing fracion o_i of tasks
- optimal outsourcing equates marginal cost of outsourcing and in-house production:

$$\underbrace{\frac{1}{z^m(o_i)}}_{\text{outsourcing cost, constant, } \uparrow \text{ in } o_i} = \underbrace{\frac{c^v \cdot x_i(o_i)^{\frac{1-\eta}{\eta}}}{\eta}}_{\text{in-house cost, } \uparrow \text{ in scale}}$$

- larger (and growing!) firms have higher RTS and higher θ^M, lower θ^L, θ^K

$\Rightarrow$ Non-homothetic prod + competitive markets vs. Homothetic prod + monopsony

Recap

- Great paper: convincing evidence on diminishing output elasticity of labor
- Key strength: unlike of most of literature, anatomy of firm-level labor share movements allowing for both technological and market power effects
- Suggestions to improve paper further
  - Use consistent class of production functions throughout
  - Ruling out non-homothetic production function and competitive markets
  - Dig into mechanism: decomposition of increased M spending can help